MAJOR PROJECT REPORT · SKILL-FIRST HIRING PLATFORM

SkillBridge

A full-stack web platform for tier-2/3 Indian college students: structured learning roadmaps, server-graded skill tests, and a job pipeline with verified recruiters — proving that what you can do matters more than where you studied.

Next.js 14 · App Router | TypeScript 5 | Tailwind CSS 3 | PostgreSQL · Neon | three.js WebGL | ₹0-cost stack

PROJECT REPORT

OCTOBER 2026

01 · INTRODUCTION

The Problem & the Idea

In India, students from tier-2 and tier-3 colleges face a structural disadvantage: recruiters filter first on college brand and pedigree, then on skills. Résumés carry no verifiable proof of ability, so capable self-taught students never reach the interview stage — while recruiters sort through hundreds of unqualified applications.

SkillBridge flips this model to skill-first: students **learn** from structured role roadmaps, **prove** their ability through timed, server-graded tests whose scores cannot be faked, and **get hired** by verified recruiters who see proof instead of pedigree. College and tier information is deliberately hidden from recruiters — performance speaks, background doesn't.

42	4	9	5
MDX LESSONS	ROLE ROADMAPS	SECURITY STEPS	DB TABLES · LIVE

Learn

42 lessons across Frontend, Backend, Data Analyst and QA roadmaps, each with an inline graded quiz.

Prove

20-minute timed tests, graded entirely server-side against a browser-invisible answer key.

Get Hired

Verified recruiters post jobs; applications carry verified skill scores automatically.

Verified everywhere

Admin verification queue for recruiters, rate-limited auth, signed test tickets.

02 · FEATURES

What the Platform Does

2.1 · Learning Roadmaps

Four complete role tracks — **Frontend Developer** (15 topics), **Backend Developer** (12), **Data Analyst** (8), and **QA / Testing** (7) — together 42 MDX-rendered lessons served as React Server Components with client JS disabled for content. Each lesson ends with an inline quiz; two project scripts (`check:lessons`, `check:links`) verify content integrity (42/42 pass) on every run.

2.2 · Server-Graded Skill Tests

Skill tests are the trust anchor of the platform. The question bank, answers, and explanations live in a module marked `server-only` — the browser receives only question stems and option labels, never answers. Grading validates an HMAC-signed test ticket bound to `{userId, skillId, iat}`, re-checks the 20-minute time window server-side, and enforces a 3-attempts-per-30-days limit with a real percentile computed against all prior attempts.

2.3 · Jobs & Application Pipeline

Recruiters post internships and full-time roles describing required skills. Students in a roadmap can apply with one click; applications arrive with the student's verified test scores already attached and a computed match-percent. Recruiters progress candidates through a five-stage pipeline: **applied** → **shortlisted** → **challenge** → **interview** → **offer** (with rejection).

2.4 · Recruiter & Admin Consoles

New recruiter accounts start **unverified** and cannot post jobs until an admin approves them through a server-driven verification queue. Admins manage this from a dedicated console; every admin and recruiter route is gated twice — once in edge middleware, once again in server layouts.

2.5 · Developer Inbox

Transactional emails (welcome, verification, application events) are rendered server-side with untrusted-input escaping and safe-CTA-link sanitisation. Without an email-provider key they persist to an `outbox` table viewable in a

dev-only inbox, so the full flow is testable end-to-end at zero cost; adding a Resend key switches delivery on without code changes.

03 · ARCHITECTURE

System Design & Stack

3.1 · Technology Choices

LAYER	TECHNOLOGY	WHY
Framework	Next.js 14.2.35 · App Router · React 18	Server Components for content-heavy pages; a single codebase for UI, API routes and auth.
Language	TypeScript 5 (strict)	Type-safe contracts between client, server and database layers.
Styling	Tailwind CSS 3 + custom design system	Nomu-inspired brand: coral/cream palette, Inter + IBM Plex Mono, paper-grain texture.
Database	PostgreSQL 16 on Neon (serverless, pooled)	Free-tier serverless Postgres; lazy idempotent schema bootstrap on first use.
Auth	scrypt hashing · HMAC session cookies · Web Crypto middleware	No third-party auth dependency; tamper-proof signed sessions verified at the edge.
3D	three.js + @react-three/fiber	Animated WebGL hero demonstrating interactive graphics in the landing page.
Lessons	next-mdx-remote v6 (RSC)	Static MDX rendering with zero client JS shipped for lesson bodies.

3.2 · Data Model (5 Tables)

TABLE	PURPOSE	KEY COLUMNS
users	All roles (student / recruiter / admin)	id (PK), email (unique), password_hash (scrypt), role, verified_recruiter
attempts	Skill-test submissions	id (PK), user_id (FK → users), skill_id, score (0–100), percentile, passed, integrity_flags
applications	Job applications	id (PK), job_id, student_id (FK → users), scores (jsonb), match_percent, stage (6-state check)
posted_jobs	Verified-recruiter jobs	id (PK), recruiter_id (FK → users), role, company, skills (jsonb), type (check)
outbox	Email log / delivery queue	id (PK), to_email, subject, html, kind, delivered ('resend' | 'outbox')

The schema is idempotent (`CREATE TABLE IF NOT EXISTS` plus indexes) and applied automatically on first database use or manually via `npm run db:setup` , which also migrates any legacy JSON-file data into Postgres.

3.3 · Dual-Backend Design

Every data store branches on one environment flag (`DATABASE_URL`): with it, all persistence goes to Postgres; without it, the same interfaces fall back to JSON files. This let the whole product be built and demoed offline first, then migrated to a real database with **zero application-layer changes** — only the driver swapped underneath.

> **Migration verified:** JSON → Neon migration moved all rows (users, attempts, applications, jobs, outbox) with counts confirmed before and after, followed by a full end-to-end run against the live database: login 200, questions 200 with zero answer leakage, grading 200 (score 100 persisted and confirmed row-by-row in Neon), signup 201.

04 · SECURITY

Hardening, Verified Live

Nine security passes were implemented and each control was verified against the running application with a curl battery:

- → **Session integrity** — HMAC-signed cookie payload `{userId, role, exp}`; forged or expired tokens rejected in edge middleware via Web Crypto with constant-time comparison.
- → **Defence-in-depth roles** — middleware gates /recruiter (recruiter|admin), /admin, and admin APIs; server layouts re-check the role independently on every request.
- → **Answer-key isolation** — the test bank is `import "server-only"`; live verification confirmed the questions endpoint returns zero answers or explanations.
- → **Signed grading tickets** — grade endpoint demands a valid HMAC ticket, independently re-enforces the 20-minute + 90-second window, and caps attempts at 3 per 30 days.
- → **Rate limiting** — 5 login/signup attempts per 15 min per (IP + email); the 6th returns 429, verified.
- → **Role-abuse prevention** — signup with role=admin is rejected 403, verified.
- → **Password storage** — Node scrypt with per-user salt, never plaintext.
- → **Email safety** — HTML-escaping of user input and sanitised CTA links; site URL derived from a single config module, not request headers.
- → **Dev-tool exposure** — the developer inbox returns 404 in production; sample/demo content and fake metrics are gated to development mode.

Live-verified results: unauthenticated hits redirect (401/403/407), 6th rapid login → 429, admin self-signup → 403, grading without a ticket → 400. Honest-data policy: landing-page statistics come only from real data sources; recruiters never see college or tier.

05 · TESTING & RESULTS

Verification & Outcomes

5.1 · How It Was Tested

CHECK	METHOD	RESULT
Content integrity	`npm run check:lessons` · `check:links`	42/42 lessons pass, all roadmap links resolve
End-to-end flow	curl battery vs live Neon DB	login 200 → questions 200 (no leaks) → grade 200 → signup 201 → outbox grew
Database migration	Row counts before/after + spot reads	All rows migrated, zero data loss
Security battery	9-step control verification	All gates redirect/429/403/400 as designed
Visual/3D	WebGL console check	three.js hero renders, 0 GL errors

5.2 · Current Limitations

- → Email delivery awaits a `RESEND_API_KEY` — messages persist to the outbox meanwhile.
- → Auth currently uses custom cookie sessions; OAuth (Google) sign-in is a planned extension.
- → The recruiter marketplace is seeded-data-level, not yet production-populated with real employers.

5.3 · Future Scope

- → **Coding challenges** — in-browser compiler for hands-on role tests beyond MCQs.
- → **Interview practice** — recorded mock-interview scoring and feedback.
- → **Company-side dashboard analytics** — funnel metrics per job posting.

- → **PWA + vernacular support** — offline-capable lessons in regional languages.

06 · CONCLUSION

Delivered & What's Next

SkillBridge demonstrates that a single full-stack platform — learning content, anti-cheat assessment, and a hiring marketplace — can be built and run entirely on free infrastructure, with real database-backed state, verified security controls, and honest, non-inflated metrics. The architecture separates trust-critical logic (grading, auth, roles) into server-only layers, proving the design scales beyond a demo into a deployable product.

Delivered

Full platform: roadmaps, server-graded tests, jobs pipeline, recruiter/admin consoles, email system, live Neon Postgres backend.

Skills demonstrated

Next.js App Router & RSC, TypeScript, Postgres schema & migration design, auth & crypto, security hardening, WebGL, zero-cost DevOps.

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